



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pathology of Skeletal Muscle

Principles and Practice of Osteopathic Medicine I

William Zaloga, D.O.
Department of Clinical Specialties
wzaloga@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

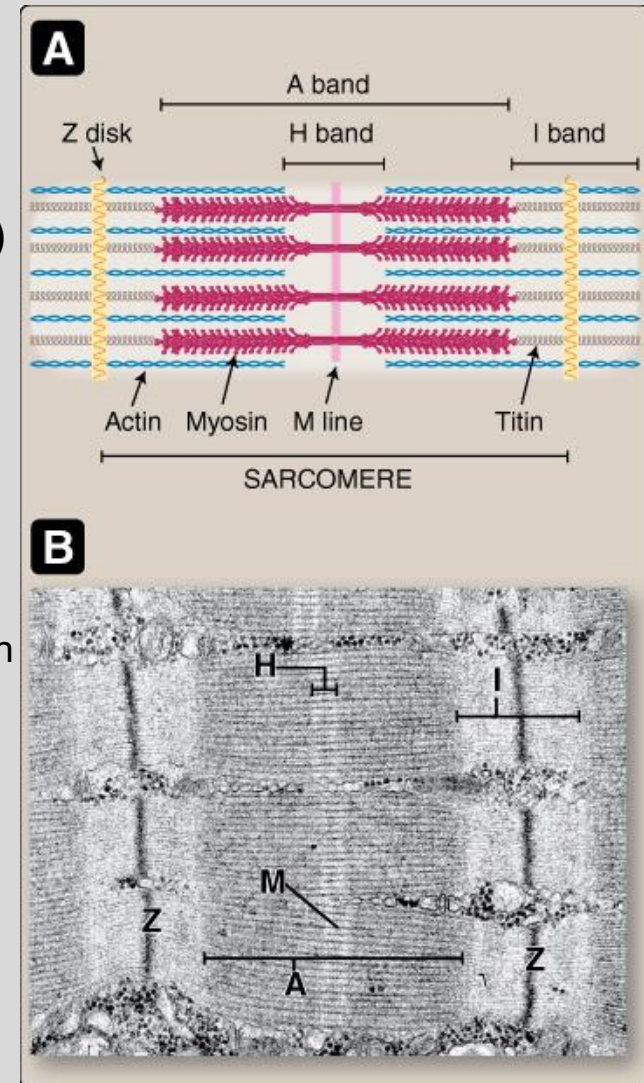
- 1. Brief review of normal muscle and, inflammatory and non-inflammatory muscle reactions and disorders such as degeneration, atrophy, etc.
- 2. Describe, discuss, and compare the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of **Spinal Muscle Atrophy (Werdnig-Hoffman) disease, Muscular Dystrophies, and Myotonic dystrophy**
- 3. Describe and discuss the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of **malignant hyperthermia (ion-channel myopathy), congenital myopathies, disorders of lipid/glycogen metabolism and mitochondrial myopathies.**
- 4. Describe, discuss and compare the pathology/pathogenesis including etiology, morphology, and clinicopathological manifestations of the major inflammatory myopathies which include **dermatomyositis, polymyositis, and inclusion body myositis.**
- 5. Describe and discuss the pathology/pathogenesis including etiology, morphology and clinicopathological manifestations of **myasthenia gravis and Lambert-Eaton myasthenic syndrome**

Skeletal Muscle Pathology

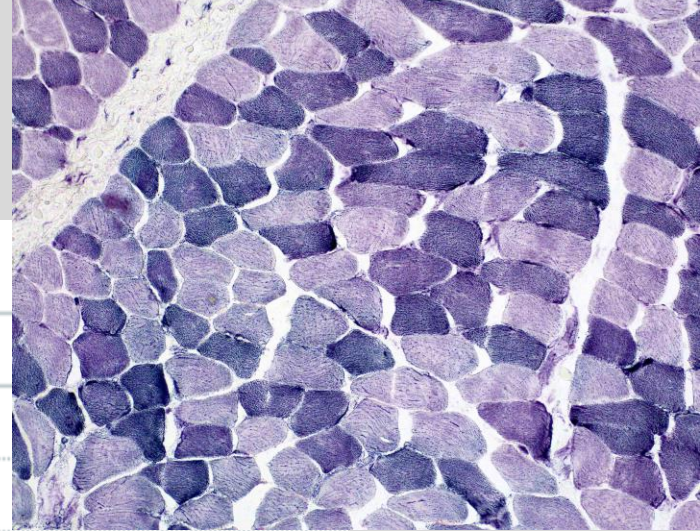
- Brief review of normal skeletal muscle
- Muscle reactions/injury
- Inherited Diseases of Skeletal Muscle
 - Spinal muscular atrophy (Werdnig-Hoffman disease)
 - Muscular Dystrophy (Duchenne and Becker)
 - Myotonic dystrophy
 - Malignant hyperthermia (Ion channel myopathy)
 - Congenital myopathies
 - Diseases of Glycogen/Lipid Metabolism
 - Mitochondrial myopathies
- Inflammatory myopathies (polymyositis, dermatomyositis, inclusion body myositis)
- Myasthenia Gravis and Lambert-Eaton Syndrome
- Miscellaneous myopathies

Normal Muscle Structure

- Sarcomere (z line to z line)
 - I band (actin filament alone)
 - A band (myosin and actin filament overlap area)
 - H band (myosin filament alone)
 - M line (attachment point of thick filaments)
- Types of Myofibers
 - Type 1: example, soleus/paraspinal muscles
 - Slow, red fibers with high levels of mitochondria
 - Sustained contractions through oxidative phosphorylation
 - Utilized in endurance training
 - Type 2: example, triceps muscle
 - Fast, white fibers with low levels of mitochondria
 - Quickly fatigue due to anaerobic glycolysis
 - Utilized in weight/resistance training or sprinting



Normal

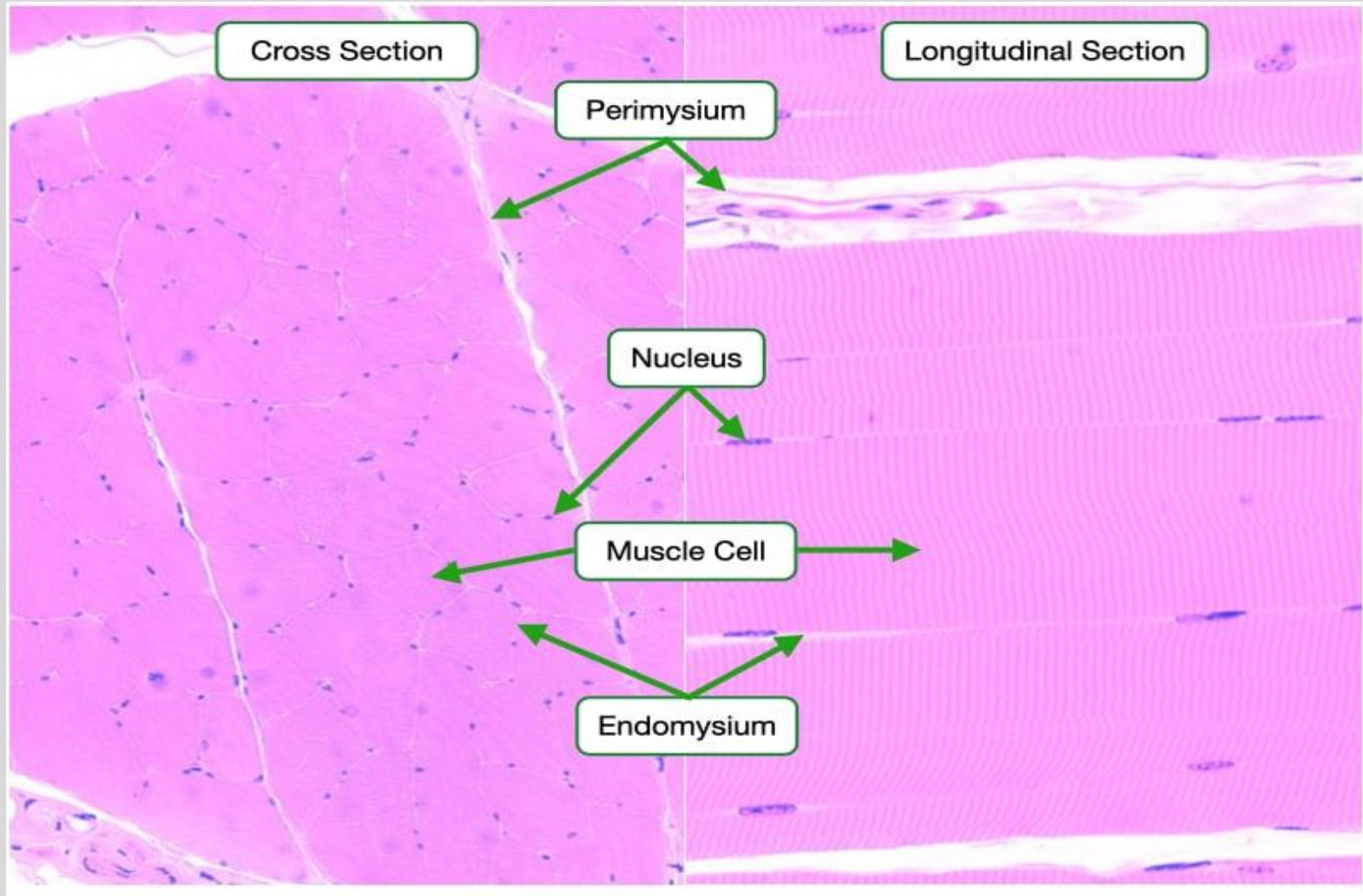


Types of muscle fibers

	Type I	Type II
CONTRACTION VELOCITY	Slow	Fast
FIBER COLOR	Red	White
PREDOMINANT METABOLISM	Oxidative phosphorylation → sustained contraction	Anaerobic glycolysis
MITOCHONDRIA, MYOGLOBIN	↑	↓
TYPE OF TRAINING	Endurance training	Weight/resistance training, sprinting
NOTES	Think “1 slow red ox”	

Normal: fibers of adjacent motor units overlap, intermingle → checkerboard pattern using nicotinamide adenine dinucleotide tetrazolium reductase (NADH-TR) stain of a snap-frozen tissue sample/flash-frozen, samples are lowered to below -70°C very rapidly using dry ice or liquid nitrogen (intensity shows proportional number of mitochondria and NADH activity, type I oxidative fibers (dark, dense purple appearance), type II nonoxidative fibers (light, scattered purple speckles))

Images: Left: First Aid, 2020, Right: <https://www.pathologyoutlines.com/topic/NADH.html>

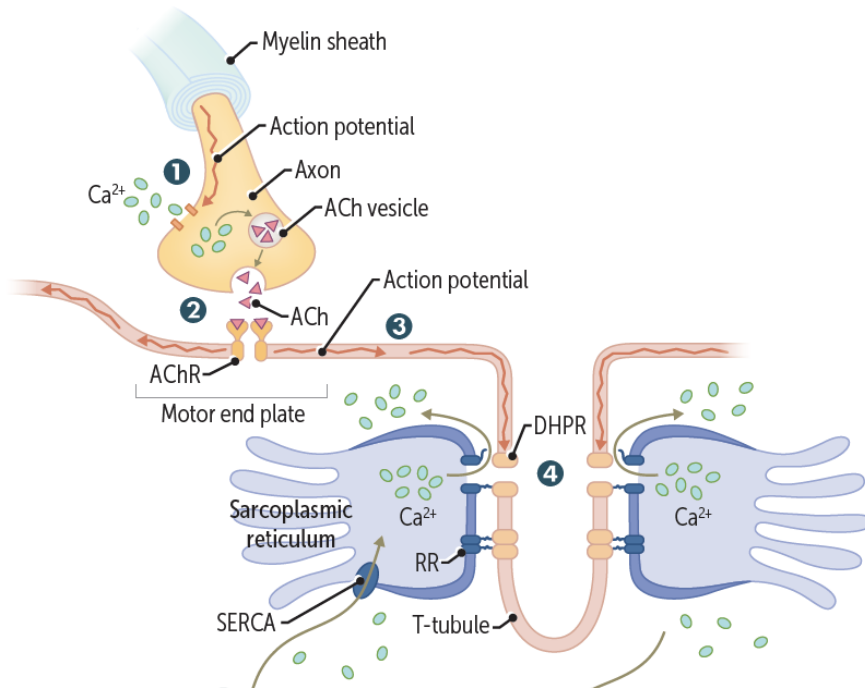


Cross section (left): skeletal muscle cells appear as large cells with peripheral nuclei

Longitudinal section (right): striations (numerous sarcomeres) identify skeletal muscle

http://medcell.org/tbl/structure_and_function_of_muscle_and_nerves/reading.php

Muscle Contraction



(1) Action potential opens voltage-gated calcium channels on the presynaptic membrane

(2) Acetylcholine (ACh) is released and binds to the ACh receptors on the postsynaptic membrane causing

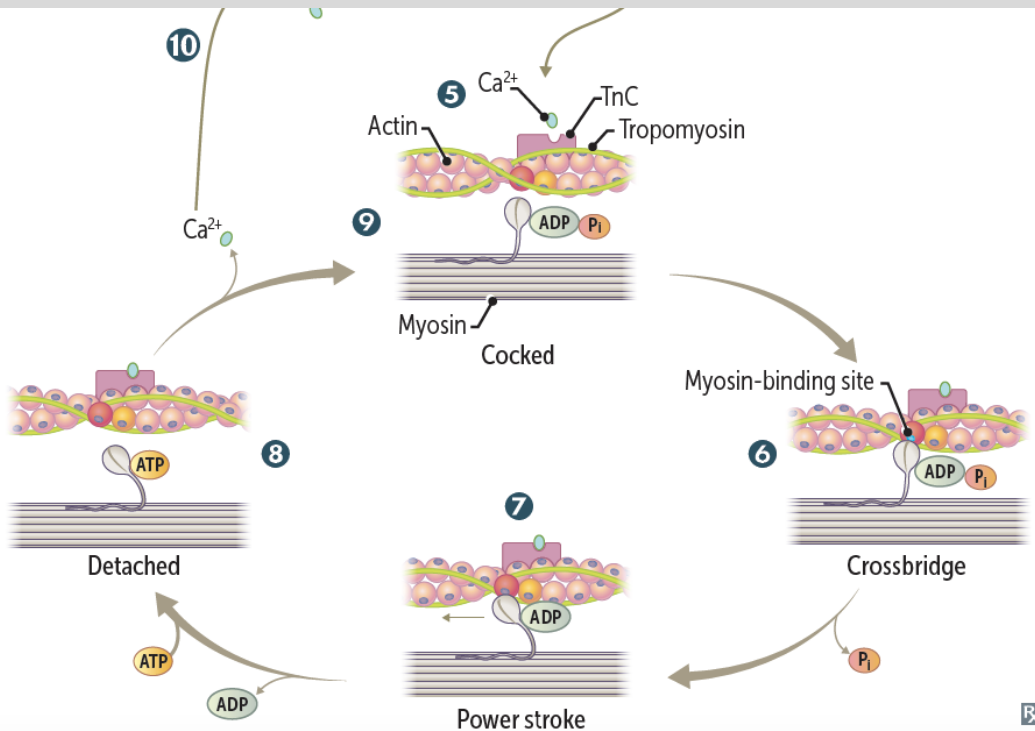
(3) Motor end plate depolarization, T-tubules connect neighboring muscle cells to coordinate contraction

(4) Depolarization opens the dihydropyridine and ryanodine receptors that allow for calcium release from the sarcoplasmic reticulum

First Aid, 2021

Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

Muscle Contraction



- (5)** Calcium binds to troponin to shift tropomyosin and
- (6)** expose the myosin-binding site on the actin filament
- (7)** Myosin head binds to actin to initiate the power stroke
- Muscle shortening occurs by shortening the H and I bands between the Z lines (A band always stays the same length)
- (8)** ATP binding releases the myosin from actin filaments,
- (9)** returning to the cocked position
- (10)** Calcium is resealed into the sarcoplasmic reticulum

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Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

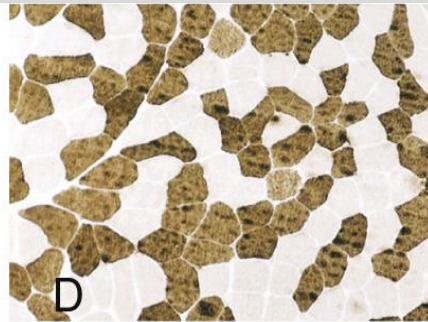
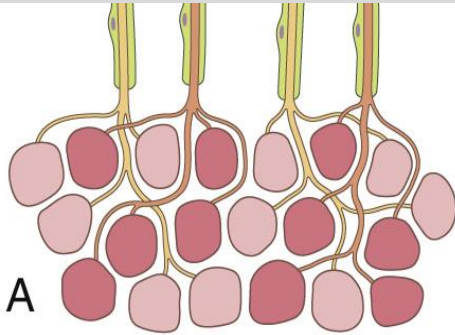
Myogenic Injury

- Segmental Myofiber Degeneration and Regeneration
 - Occurs when **part of a myofiber undergoes necrosis**
 - Degeneration
 - **Creatine kinase released into blood from damaged cells**
 - Myophagocytosis occurs (macrophages remove damaged components of the myofiber)
 - Regeneration
 - Begins with fusion of an **activated satellite cell** to the damaged myofiber
 - **Regenerated myofibers stain basophilic** due to high levels of RNA
 - In chronic disease, complete regeneration is not possible due to the **disease process outpacing regeneration**
- Myofiber hypertrophy is typically a **physiologic response to exercise** but can be associated with a disease processes
- Cytoplasmic inclusions include vacuoles, protein aggregates, clustered organelles

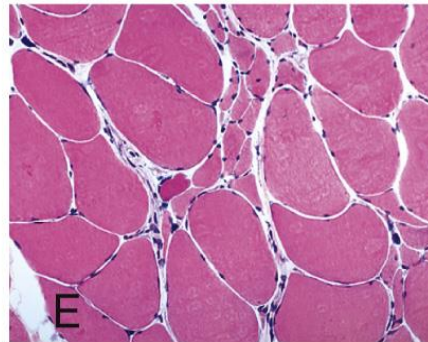
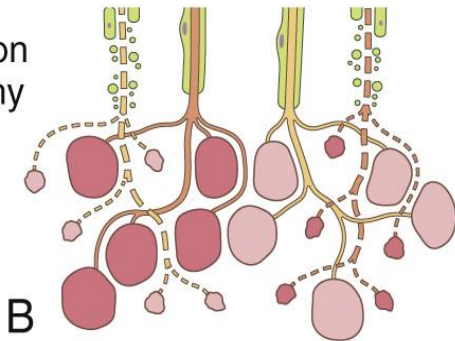
Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

Neurogenic Injuries

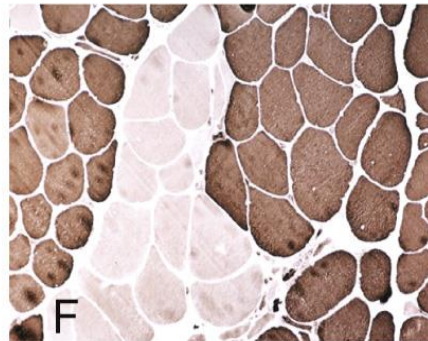
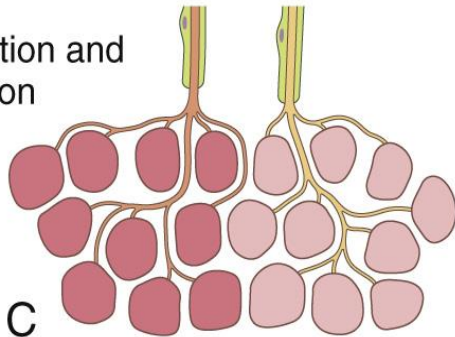
Normal



Denervation and atrophy



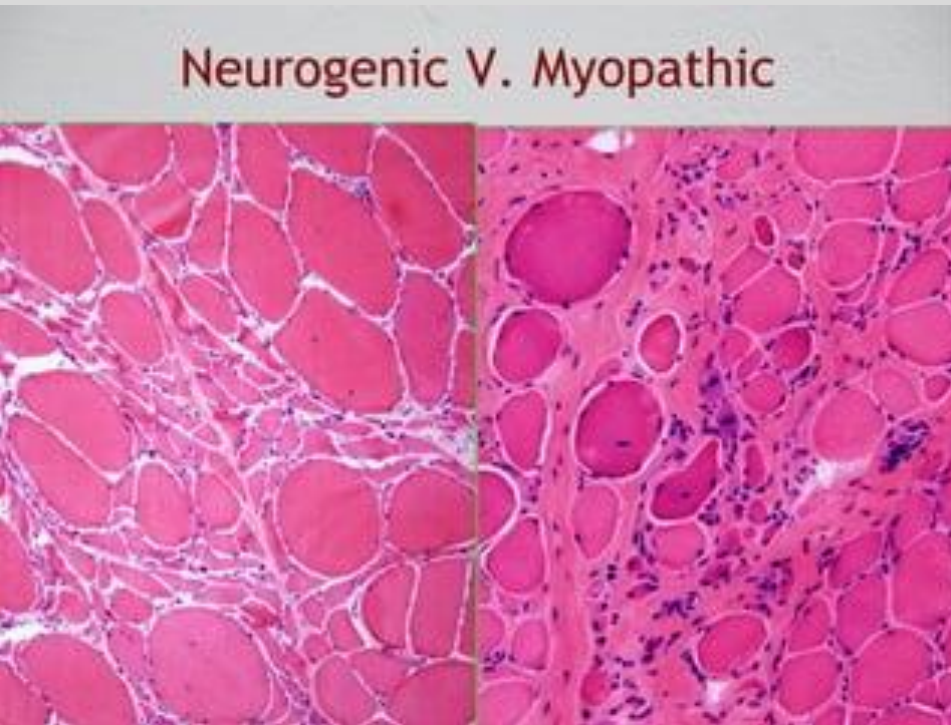
Reinnervation and regeneration



- **Disruption of innervation** of a muscle that causes muscle dysfunction (usually occurs as **grouped atrophy, E**)
- Mechanism: **innervating motor neuron determines fiber type**
 - Denervation causes **atrophy** of those fibers as a **group** (**flattened, angular shape**)
 - Reinnervation will normalize the shape but may **change the fiber type**; axons belonging to unaffected nearby motor unit extend sprouts to reinnervate myocytes; newly reinnervated fibers become fiber type of that motor unit: loss of checkerboard pattern, D (histologically motor units appear as groups of myofibers of the same histochemical type (fiber type grouping, F))

Neurogenic vs myopathic injury

- Neurogenic atrophy of muscle
 - Caused by disorders that affect motor neurons
 - Motor neuron diseases are progressive, destroy anterior horn cells in spinal cord and cranial nerve motor neurons
 - Denervation causes atrophy of those fibers as a **group** (**flattened, angular** shape)
- Myopathic disorder
 - Single fiber necrosis, and regeneration
 - **Necrotic** cells **lose stain** of nuclei with **pale** cytoplasm, **shrink**, irregular cell membrane, also cytoplasm irregular stain with **condensed** areas
 - Increased variability in fiber size, including hypertrophy and/or atrophy
 - **Regenerating** myofiber show increased **central nucleation**, more **basophilic** initially on H&E then enlarge, **hyperchromatic** cytoplasm, then nuclei **peripheral** cytoplasm less hyperchromatic



Robbins and Cotran Pathologic Basis of Disease,
9th ed. 2014, Ch. 27

Image:

<https://www.slideshare.net/gliageek/pathology-of-skeletal-muscle-presentation>

Inherited Diseases of Skeletal Muscle

- Spinal Muscular atrophy
- Muscular Dystrophy
- Myotonic Dystrophy
- Malignant Hyperthermia
- Congenital myopathies
- Diseases of Glycogen/Lipid Metabolism
- Mitochondrial Myopathies

Spinal Muscular Atrophy: SMA I

- Disorders with childhood onset, progressive weakness
 - **SMA I = Werdnig-Hoffman's Disease**; most severe
- Pathogenesis: **Autosomal Recessive**: loss of fxn mutation in SMN1 gene (chrom 5) [possible role in RNA splicing (snRNP assembly)] – causes defects in spliceosomes
 - **Loss of motor neuron innervation (anterior horn cell destruction/degeneration) → muscle weakness and atrophy**
 - **Degeneration of the anterior horns (motor neurons) of the spinal cord**
 - **Denervation of skeletal muscle**
- Clinical features –LMN signs
 - Infants: can present with **generalized hypotonia (“floppy infant”)-little muscle tone, presents within 4 months of birth; death usually by 2 - 3 years**
- Diagnostic features - biopsy
 - Histology: large areas of flattened, angulated, very atrophic myofibers mixed with normal/hypertrophied myofibers in small groups (those that retain innervation)

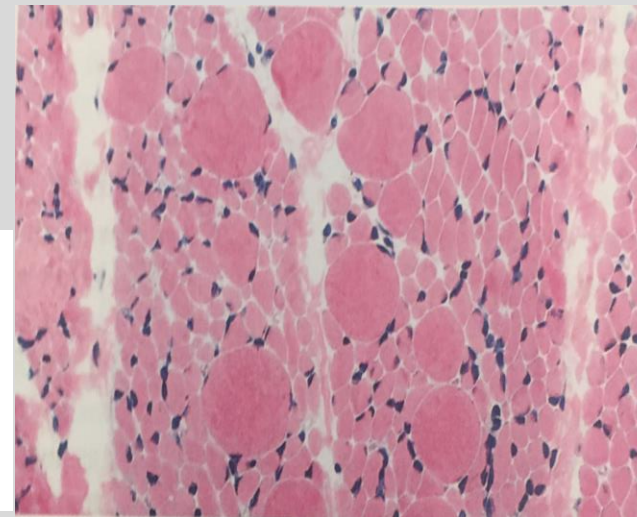
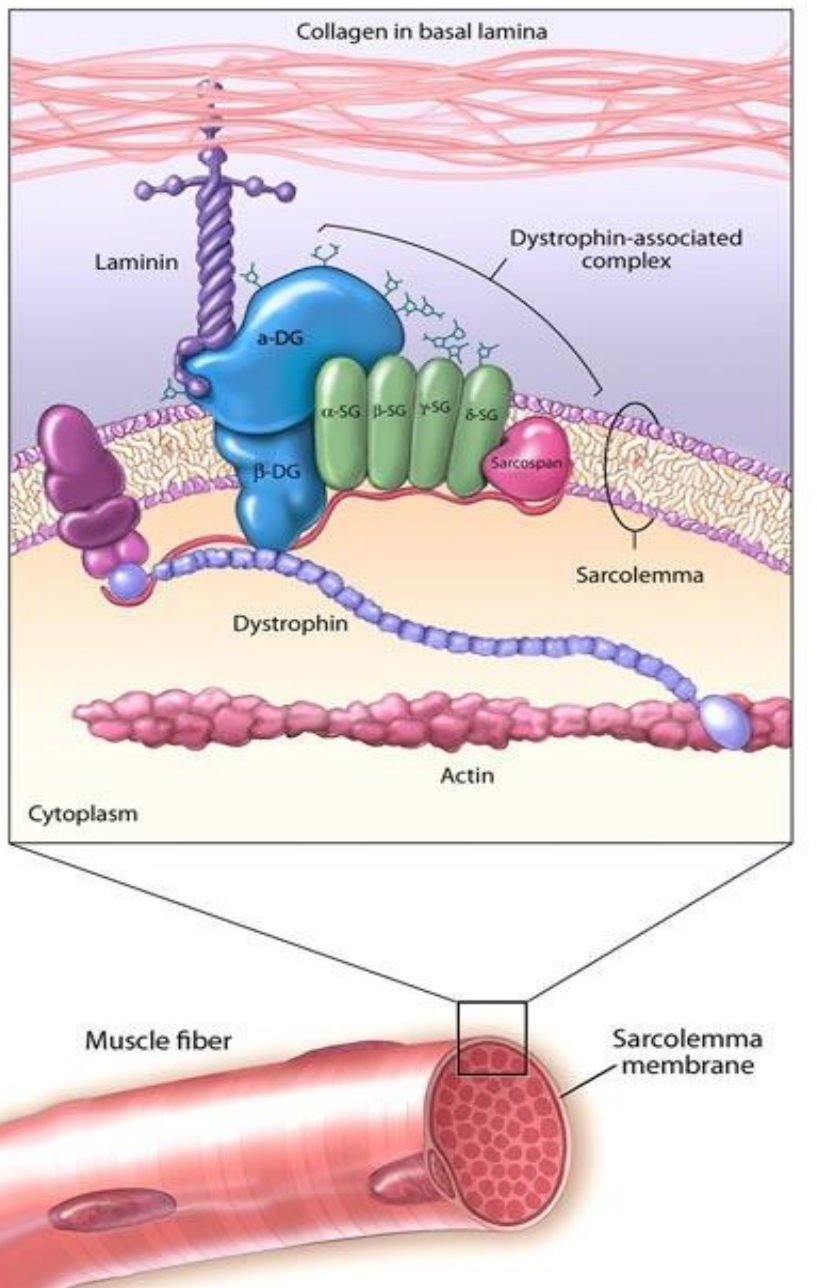


Figure 27. 13 Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27
First Aid 13th ed. 2020

Muscular Dystrophies

- Progressive muscle weakness and wasting, usually beginning in early childhood
- Two most common: Duchenne MD and Becker MD
- X-linked – gene located in Xp21 region, encodes **dystrophin**
- Duchenne - more severe, usually leads to death by 20s – 30s.
- Becker – milder phenotype, later onset of disease

Robbins and Cotran Pathologic Basis of Disease, 9th ed. 2014, Ch. 27

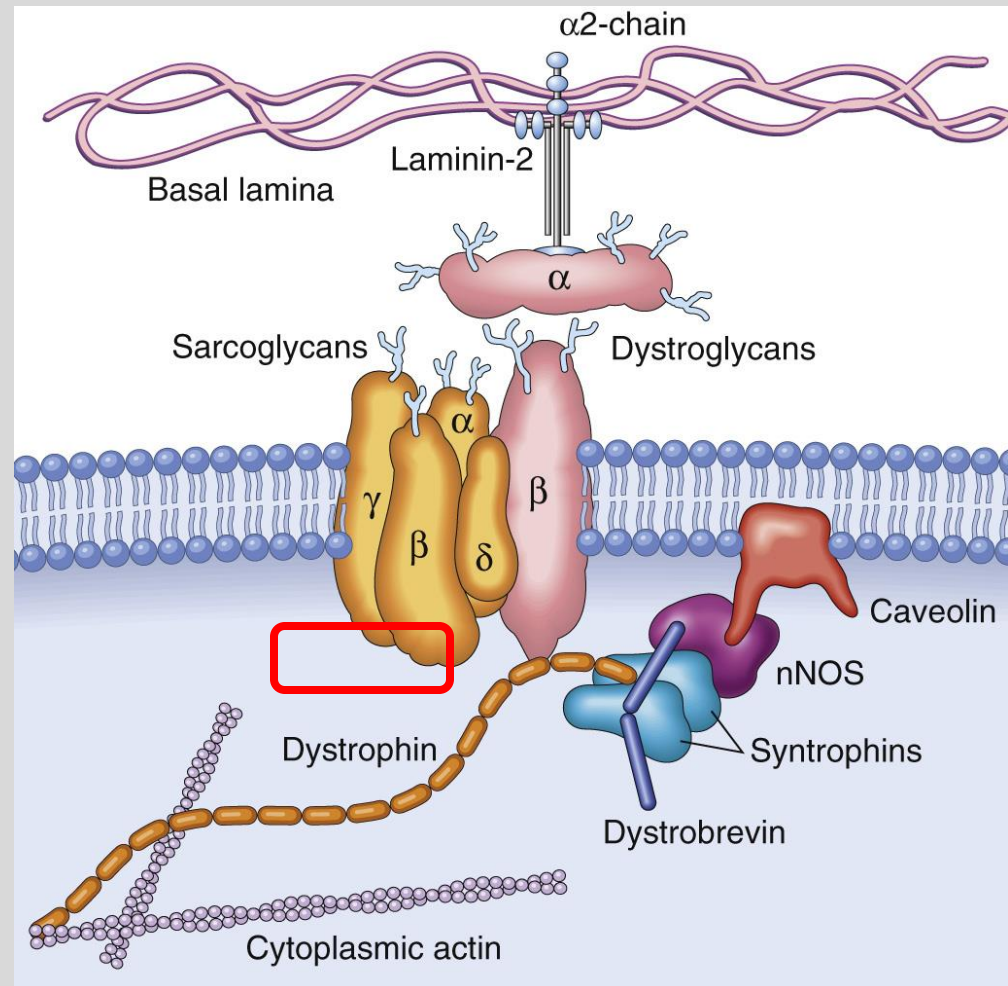


Dystrophin – cytoplasmic protein next to sarcolemmal membrane

- Concentrates at plasma membrane over Z band, forms **strong link to actin**
- **Helps connect cytoskeleton with ECM (mechanical stability):**
Dystrophin and dystrophin-associated complex form interface between intracellular contractile apparatus and extracellular connective tissue matrix and transfer of force of contraction to connective tissue
- **Myocyte degeneration:** When dystrophin is absent or abnormal progressive **myocyte damage occurs over time that exceeds regenerative capacity**

Duchenne Muscular Dystrophy (DMD)

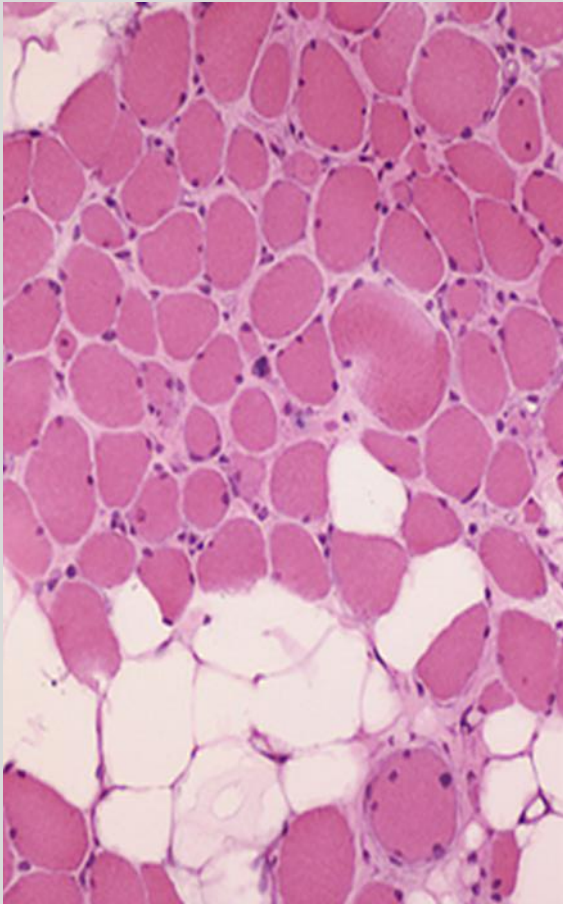
- Pathogenesis: **X-linked recessive**, frameshift deletion or nonsense mutation of the dystrophin gene
 - **Absent** dystrophin protein
 - Chronic muscle damage that cannot be contained by the repair mechanisms



DMD Histopathology

- **Early disease:**

- muscle biopsies in young boys: ongoing muscle damage, i.e., **segmental myofiber degeneration and regeneration** associated with admixture of **atrophic myofibers**
- **myophagocytosis, usually no inflammation**
- **fascicular architecture still preserved**

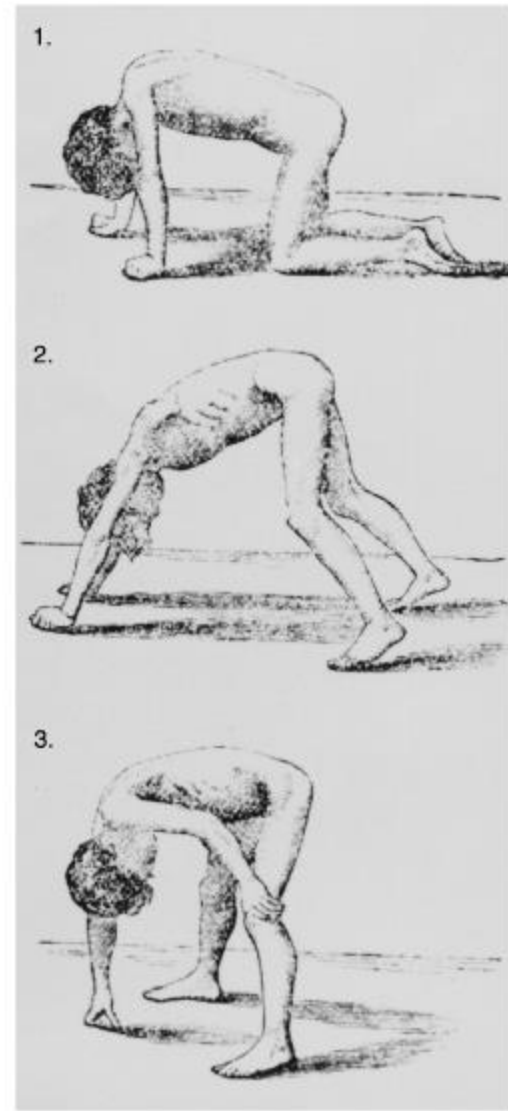


- **As DMD progresses:**

- muscle tissue **replaced by collagen and fat cells** (“fatty infiltration”, image) - causes pseudohypertrophy of calves
- remaining myofibers: **prominent variation in size** (small atrophic fibers to large hypertrophied fibers)
- **fascicular architecture distorted, abnormal**

DMD, Gowers maneuver

- Duchenne MD manifested by age 5 years - Walking may be delayed
- Life expectancy 25-30 years
- Weakness in pelvic girdle muscles, extends to shoulder muscles
- Enlargement of calf muscles associated with weakness – pseudohypertrophy (increase in muscle fibers due to increase in fat and fibroconnective tissue- as muscle atrophies)



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Drawing of a boy with Duchenne muscular dystrophy rising from the ground (the **Gowers** maneuver). (From Gowers WR [1879] Pseudohypertrophic Muscular Paralysis. A Clinical Lecture. J. and A. Churchill, London.) Maneuver used by patients with muscle weakness to arise from a supine or seated position on the floor. Patients push with their hands to "climb up" their legs.

DMD

- Clinical features: Usually **Proximal muscle weakness and clumsiness**
 - **Pseudohypertrophy of the calf muscles**: enlargement of the muscle with associated weakness (**muscles are replaced by collagen and fat tissue**)
 - **Gower's Sign**: use of their hands to stand from a lying/sitting position
 - Later, develop joint contractures, scoliosis, worsening respiratory reserve and sleep hypoventilation
 - Can develop cardiomyopathy and arrhythmias (protein is located primarily in skeletal muscles and in heart, small amounts of dystrophin are present in nerve cells in brain)
 - May have intellectual impairment
 - Usually increased serum creatine kinase (early stages)
 - **Death may result from respiratory insufficiency by the 20s – 30s**
- Treatment: supportive; New efforts to manipulate RNA splicing, promotion of ribosomal “read-through” of stop codons

Becker Muscular Dystrophy

- Pathogenesis: **X-linked recessive, loss-of-function** mutation of the **dystrophin** gene
 - Results in a truncated, **partially functional** dystrophin protein
- Presents in **adolescence or adulthood**
- Clinical features:
 - Similar to Duchenne but **less severe and slower progression**
 - Presents later
 - May have normal life expectancy
- Treatment: supportive care

Question

- A 4 year old boy is examined by a physician because his parents complain he falls a lot, cannot jump, and tires easily. PE shows weakness in the pelvic and shoulder girdles and enlargement of the calf muscles. The serum creatine kinase is elevated. A biopsy of calf muscle is performed and shows marked variation in size and shape of muscle fibers. There are also foci of muscle fiber necrosis, myophagocytosis, regenerating fibers, and fibrosis. What would most likely may be the eventual cause of death in this patient?
- A. Ruptured brain aneurysm
- B. Ruptured aortic aneurysm
- C. Respiratory insufficiency
- D. Liver failure
- E. Kidney failure

Answer

- A 4 year old boy is examined by a physician because his parents complain he falls a lot, cannot jump, and tires easily. PE shows weakness in the pelvic and shoulder girdles and enlargement of the calf muscles. The serum creatine kinase is elevated. A biopsy of calf muscle is performed and shows marked variation in size and shape of muscle fibers. There are also foci of muscle fiber necrosis, myophagocytosis, regenerating fibers, and fibrosis. What would most likely may be the eventual cause of death in this patient?
- A. Ruptured brain aneurysm
- B. Ruptured aortic aneurysm
- **C. Respiratory insufficiency (this and cardiomyopathy most commonly)**
- D. Liver failure
- E. Kidney failure

Myotonic Dystrophy

- Pathogenesis: **Autosomal dominant, CTG trinucleotide repeat**
- Age of onset **20s-40 years**
- Length of expansion correlates with disease severity (similar to Huntington disease)
- Triplet expansion disrupts normal splicing
 - Repeat is expanded in the **DMPK gene** that encodes for **myotonin protein kinase**
 - Disruption of RNA splicing causes **deficiency of CLC1 (chloride channel)** normally involved with muscle **relaxation**: leading to **myotonia**

Myotonic Dystrophy

- **Myotonia** - abnormally slow relaxation of muscles; prolonged contractions (<https://youtu.be/Wg1SVoa-8JE>) with
 - Progressive muscle wasting and weakness (type 1 **fiber atrophy**)
 - **Grip myotonia** (cannot relax, sustained grip)
 - Facial muscle weakness
 - Frontal balding
 - Cataracts
 - Testicular atrophy
 - Glucose intolerance
 - Conduction defects in heart (arrhythmias) and cardiomyopathy
 - Increased serum CPK (like muscular dystrophy)

Robbins and Cotran Pathologic Basis of Disease, 9th ed. 2014, Ch. 27

Ion Channel Myopathies

- These **skeletal muscle channelopathies** are inherited mutations affecting function of ion channel proteins
 - Some increase excitability (hypertonia), some decreased excitability (hypotonia)
 - Some are subclassified based on K levels (examples, hyperkalemic, or hypokalemic, or normokalemic periodic paralysis)
 - Depending on affected ion channel, clinical manifestations may include:
 - epilepsy, migraines
 - movement disorders with cerebellar dysfunction
 - peripheral nerve disease, muscle disease

Ion Channel Myopathies cont.

- Some examples of mutations assoc. with muscle dysfunction:
- CACNA1S protein is subunit of muscle calcium channel and a missense mutation most common cause of **hypokalemic paralysis**
- CLC1 mutation affects chloride channel, associated with myotonia congenita, and **CLC1 expression decreased in myotonic dystrophy**
- **Mutations RYR1 → disrupt function of ryanodine receptor (regulates calcium release from the sarcoplasmic reticulum) and are involved in congenital myopathy (central core disease) and malignant hyperthermia**

Malignant Hyperthermia

- Rare but life-threatening **skeletal muscle channelopathies**
Pathogenesis: **Autosomal dominant: Mutation in the voltage-gated RYR1 gene affects ryanodine-r (involved with controlling release of Ca²⁺ from sarcoplasmic reticulum)**
 - **Increased calcium release** from sarcoplasmic reticulum following administration of **halogenated inhaled anesthetics and succinylcholine (temporary paralysis, rapid onset and short duration of muscle relaxation)**
 - **Mechanism: Interaction of anesthetic and mutated receptor causes incr. calcium efflux which leads to tetany and heat production**

Malignant Hyperthermia clinical features

- **Hypermetabolic state**
 - **Hyperthermia** and later, hyperpyrexia (temp > 106 degrees F)
 - Severe muscle contractions (**tetany**)
 - **Tachycardia/Tachypnea**
- May occur in predisposed patients with **hereditary muscle disease**, inc. congenital myopathies, dystrophies, and metabolic myopathies
- Treatment: **Dantrolene** (ryanodine receptor antagonist)

Question

- A 25 yo male is in the OR for appendectomy. A few minutes after receiving succinylcholine, his vitals appear as follows: BP 150/100, HR 120 BPM, RR 22, Temp 102.8 F. A mutation in which of the following is likely responsible for this patient's presentation?
- A. SMN1
- B. RYR1
- C. DMPK
- D. BCL-2

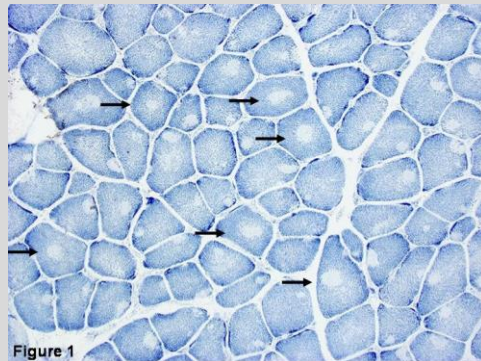
Answer

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- A. SMN1
- **B. RYR1**
- C. DMPK
- D. BCL-2

Congenital Myopathies

Central Core Disease

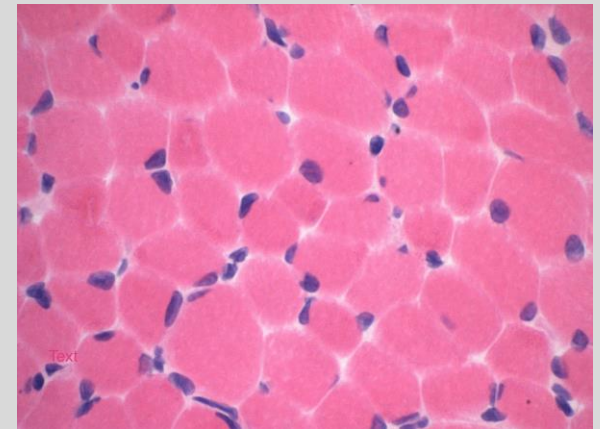
- Pathogenesis: Autosomal Dominant
 - **Mutation of the ryanodine receptor (RYR1) gene**
- Clinical features: Usually hypotonia at birth
 - **Associated with skeletal abnormalities (scoliosis), may develop malignant hyperthermia**
- Diagnosis:
 - Biopsy: central/eccentric zones of disrupted sarcomere arrangement (areas of reduced oxidative and glycolytic enzymatic activity)



NADH-TR stain, transverse section from the rectus femoris
Jungbluth H. Central core disease. Orphanet Journal of Rare Diseases. 2, 25.
2007. doi:10.1186/1750-1172-2-25

Nemaline Myopathy

- Pathogenesis: Autosomal dominant or recessive, mutations involving **thin filament** of sarcomere and cause contractile dysfunction, leading to muscle weakness
- Clinical features: Childhood weakness
 - May present at birth as “floppy” baby
- Diagnosis:
 - Biopsy: Nemaline (**thread-like**) rods (aggregates of spindle-shaped particles in type 1 fibers)



H&Estain

<https://www.pathologyoutlines.com/imgau/nemalinemyopathyHiser01.jpg>

Disorders of Lipid or Glycogen Metabolism

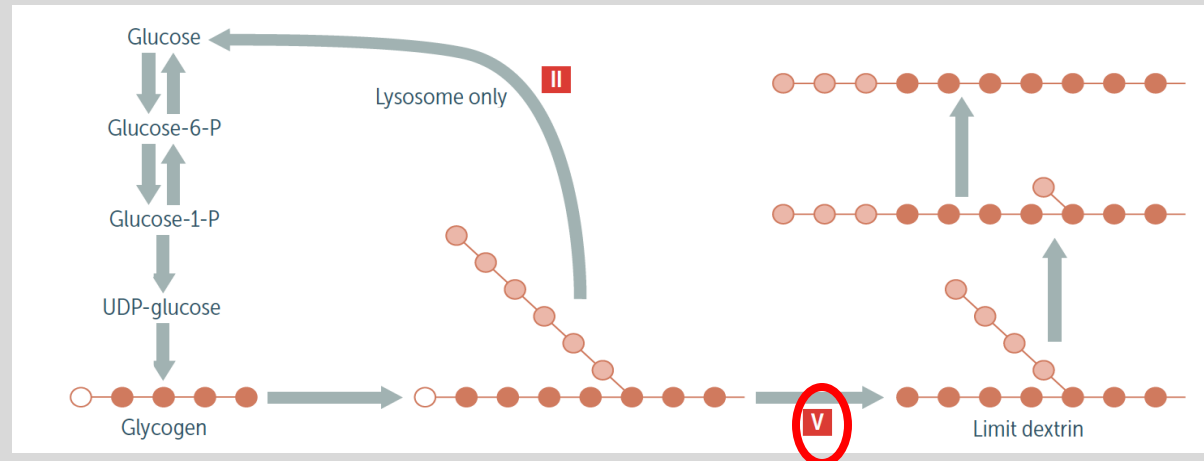
- General patterns of muscle dysfunction:
 1. Symptomatic with exercise/fasting which range from muscle cramping to rhabdomyolysis
 2. OR slowly progressive muscle damage, no episodic symptoms
- Examples (more details follow)
- **Carnitine palmitoyltransferase II deficiency**
 - Defect in free fatty acid transport into mitochondria
- **Myophosphorylase deficiency (McArdle disease)**
 - Episodic damage, eg. myoglobinuria
- **Acid maltase deficiency (Pompe disease)**
 - Impaired lysosomal conversion glycogen to glucose

Carnitine Palmitoyltransferase II Deficiency

- **Most common disorder of lipid metabolism**
- Pathogenesis: Enzyme deficiency
 - Impaired transport of free fatty acids into the mitochondria
 - **Toxic build up of fatty acids in the cytosol**
- Clinical features: Episodic muscle damage with exercise or fasting
 - **Hypotonia**
 - Hypoketotic hypoglycemia
 - Dilated cardiomyopathy

Diseases of Glycogen Metabolism

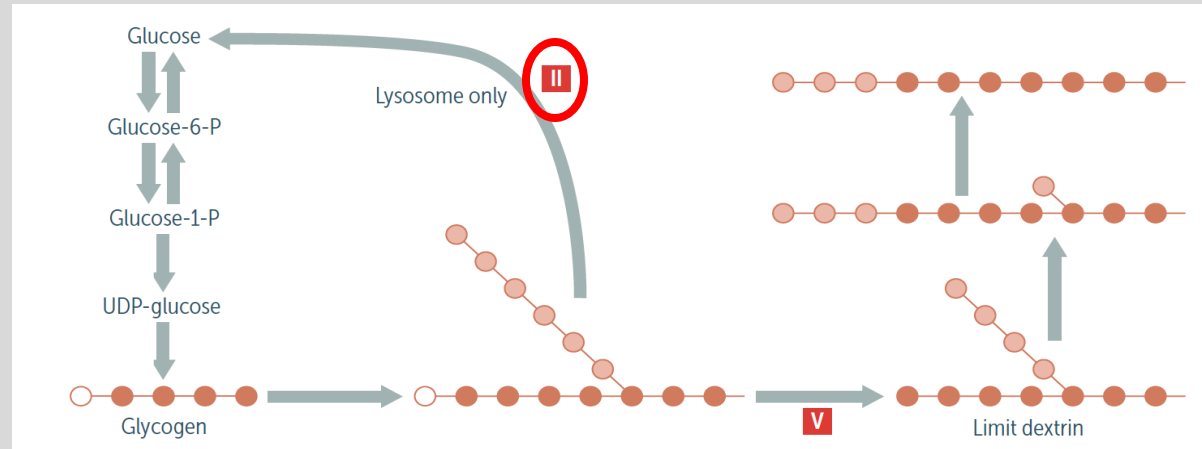
McArdle Disease



- Pathogenesis: **Myophosphorylase (skeletal muscle glycogen phosphorylase) deficiency (type V)**
 - **Inability to break down glycogen in muscle**
- Clinical features: Episodic muscle damage with exercise
 - **Painful muscle cramps**
 - **Myoglobinuria**
 - **Arrhythmia** (due to electrolyte imbalances)
 - **Normal blood glucose levels** (liver is unaffected)
- Diagnosis: Venous lactate-ammonia test, during exercise, **venous lactate curve is flat** with a normal rise in ammonia levels
- Treatment: Limit carbohydrates rich diet and careful exercise routine

Diseases of Glycogen Metabolism

Pompe Disease



- Pathogenesis: **Acid maltase (lysosomal acid α -1,4-glucosidase) deficiency (type II)**
 - Impaired glycogen breakdown within the lysosome
- Clinical features: Heart, liver and muscle can be affected
 - **Cardiomegaly** (Hypertrophic cardiomyopathy)
 - **Hypotonia** (severe deficiency presents in infancy)
 - Exercise intolerance
 - Mild deficiency affects respiratory and truncal muscles
 - Early death is seen in patients with systemic symptoms
- Treatment: Enzyme replacement therapy works for some patients

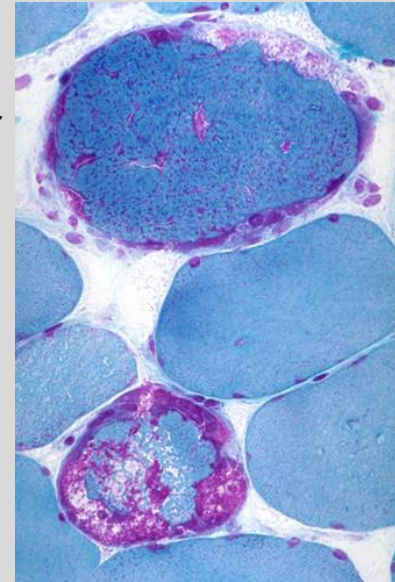
Mitochondrial Myopathies

- Rare disorders; may involve many organ systems
- Pathogenesis: Involves mutations of **mitochondrial DNA** with **maternal inheritance** because only oocyte contributes mitochondria to embryo. Disease is thought to be due to **mutations that impair ATP generation** in mitochondria
 - **All children of an affected female** may show signs of disease due to maternal transmission (affected males can't transmit to their children), has varying degrees of expressivity
 - Tend to **affect skeletal muscles/other tissues with high ATP requirements (cardiac muscle cells-dysrhythmia, neurons-seizures)**, drug therapy for MERRF syndrome includes metabolic drugs involved with or improve cellular respiration (ex. metformin, creatine, and L-carnitine), clonazepam for myoclonus and anticonvulsants
 - Skeletal muscle involvement causes weakness, increase in serum **CK**, or **rhabdomyolysis**
 - Weakness in certain regions is variable, but **involvement of extraocular eye muscles common**
 - Chronic progressive **external ophthalmoplegia** (inability to move eyes, eyelids and eyebrows – blurred, double vision) is common feature of mitochondrial disorders

Mitochondrial Myopathies

- **Myoclonic epilepsy with ragged red fibers (MERRF)**

- **Multisystem** mitochondrial diseases **characterized by predominantly intermittent muscle twitches aggravated by action (progressive myoclonus**, hiccups are a form of myoclonus, also the sudden jerks, or "sleep starts," you may feel just before falling asleep - these forms of myoclonus occur in healthy people and rarely present a problem), **weakness (myopathy – limb girdle type), and progressive stiffness (spasticity)**, diagnosis/biopsy: "ragged red fibers" due to accumulation of affected mitochondria in the subsarcolemmal area of myofibers, increased **serum creatine kinase**
- Most also experience other types of seizures in addition to myoclonus, may be of generalized tonic-clonic, atonic or absence types
- Modified **Gomori trichrome** stain allows visualization of the mitochondrial activity and collection by staining red (ragged red fibers), Pathogenic variant in mitochondrial gene MT-TK is present in more than 80% of affected



Mitochondrial clusters appear as red deposits in cryostat sections of muscle stained with modified Gomori trichrome
<https://neuropathology-web.org/chapter10/images10/10-rrfl.jpg>

- **Leber hereditary optic neuropathy**

- **Predominantly** degeneration of retinal ganglion cells and their axons leads to an acute or subacute loss of central vision; it primarily affects young adult **males**; different mitochondrial mutation from MERRF
- Commonly present with chronic progressive external **ophthalmoplegia** (weakness of the extraocular eye muscles)

Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

First Aid 13th ed. 2020

Inflammatory Myopathies

- Immune mediated – inflammation/injury (noninfectious) of skeletal muscles (muscle **weakness** and **myalgia** major symptoms), examples
 - Dermatomyositis
 - Polymyositis
 - Inclusion Body Myositis

Dermatomyositis

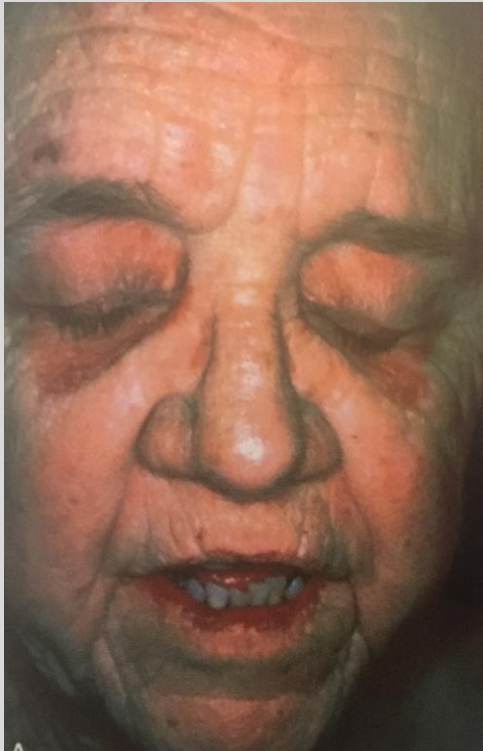
- Rare inflammation of skin and skeletal muscle; **Immune-mediated** small blood **vessel** injury leads to **muscle damage**, peak age onset **50-60 years**, more common in **females**, gradually worsens over weeks to months, can last several years or be lifelong
 - Classic skin rash: **Lilac** or **heliotrope** (reddish-purple rash on or around the eyelids) with **periorbital edema**
 - May have **Gottron's lesions/papules** (scaling erythematous eruption or dusky red patches on knuckles, elbows, or knees).
 - **Proximal muscles weak first; symmetric**, slow onset, **myalgias**
 - Dysphagia may develop if oropharyngeal, esophageal muscles involved
 - **Extramuscular involvement** includes interstitial lung disease (10%) vasculitis and myocarditis
 - 15 - 24% of adults with dermatomyositis may have **malignancy** (Lung, breast, ovary, colorectal, cervical, bladder, nasopharyngeal, esophageal, kidney, lymphatic, hematopoietic)

Dermatomyositis

- Pathogenesis: **Autoimmune** disease characterized by small vessel damage, **capillaries** attacked by Abs and complement causing foci of **ischemic myocyte necrosis**
- Although the cause of dermatomyositis is unknown, several genetic, immunologic, and environmental factors are implicated, having another autoimmune disease is a risk factor
- **Anti-Mi2 Abs** (anti-helicase) associated with **Gottron papules** and **heliotropic rash**
- **Anti-Jo1 Abs** (anti-histidyl t-RNA synthetase): associated with **interstitial lung disease, non-erosive arthritis** and **“mechanic’s hand” rash**

Dermatomyositis cont.

Heliotrope Rash



Dermatomyositis – Gottron papule



“Mechanic’s Hands”

- roughened, cracked skin at tips and lateral aspects of the fingers resulting in irregular, dirty-appearing lines



<https://image2.slideserve.com/5103882/mechanic-s-hands-l.jpg>

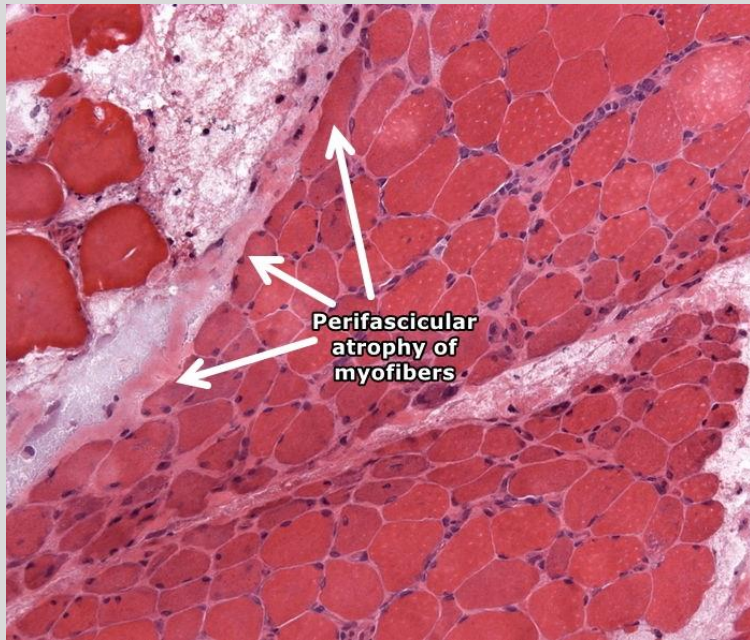
Figure 27.8 Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

<https://medlineplus.gov/ency/imagepages/1522.htm>

Ricceri, F., & Prignano, F. (2013). Gottron papules: a pathognomonic sign of dermatomyositis. *CMAJ : Canadian Medical Association journal = journal de l'Association medicale canadienne*, 185(2), 148.

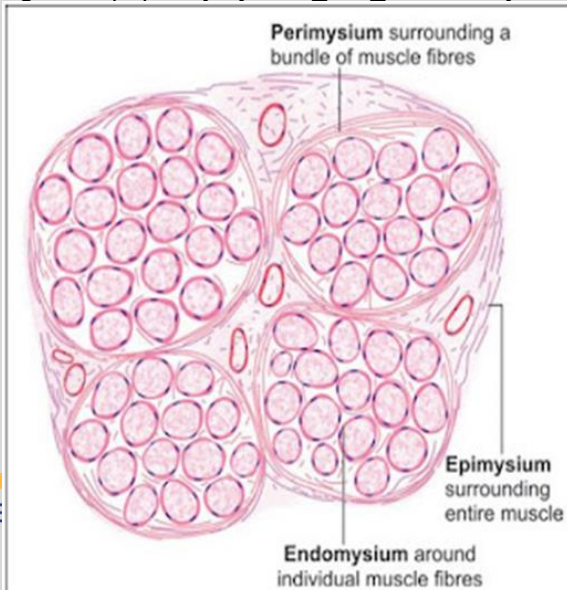
<https://doi.org/10.1503/cmaj.111791>

Dermatomyositis



- Diagnosis:
 - Biopsy: Deposition of the **complement** MAC (C5b-9) within the capillary beds
 - Histology: **Perifascicular inflammation/atrophy of myofibers and loss of vessels around muscle fibers, CD4+ T-helper cell infiltration (perimysial)**
 - Autoantibody detection
 - **Increased creatine kinase** on labs (reflects muscle damage)
- Treatment: immunosuppressive agents

https://www.wikidoc.org/index.php/Polymyositis_and_dermatomyositis_pathophysiology



First Aid, 2021

Robbins and Cotran Pathologic Basis of Disease, 10th ed. 2021, Ch. 27

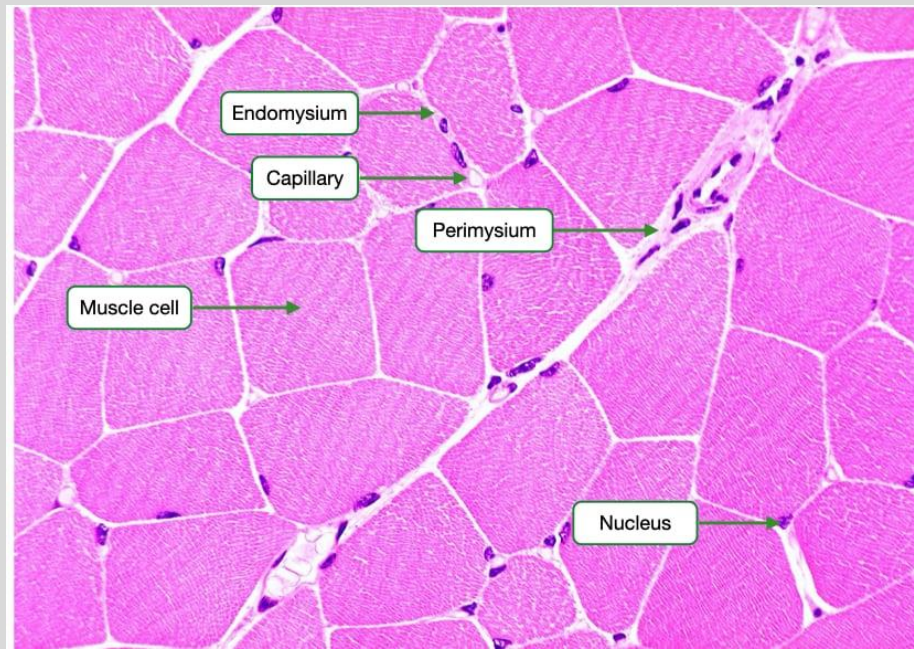
<https://www.medicinehack.com/2016/12/endomysium-perimysium-and-epimysium.html>

Polymyositis

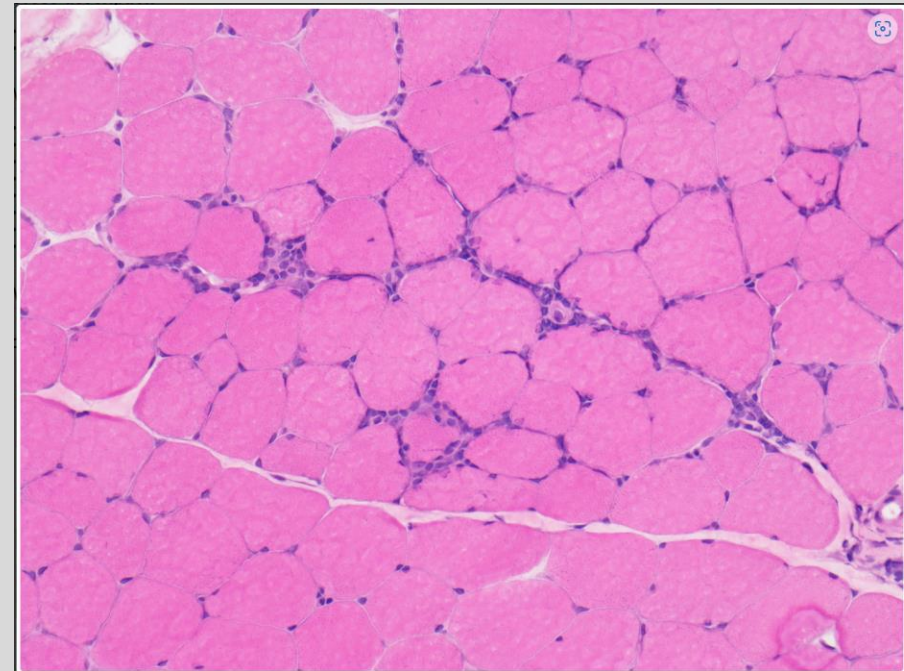
- Symmetric proximal **muscle** involvement **similar to dermatomyositis but no skin involvement**, occurs mainly in **adults**, (peak age onset 30-60 years) and may have inflammatory involvement of **heart, lungs, and blood vessels**, women are affected more often than men, can last several years or be lifelong
- Pathogenesis: Immune-mediated (**CD8+ cytotoxic T cells**) **tissue damage, there is no vascular injury** (in contrast with dermatomyositis)
- Polymyositis is not caused by a genetic mutation, but some genes may influence the risk of developing polymyositis, having another autoimmune disease is a risk factor, exact cause unknown, but has characteristics of autoimmune disorders, which occur when the immune system mistakenly attacks healthy body tissues
- Histology: **Endomysial inflammation** (predominately CD8+ T cells)
 - Necrotic and regenerating muscle fibers scattered throughout fascicle
 - **No perifascicular atrophy** (seen in dermatomyositis); no evidence of vascular damage
- Clinical Findings: **myalgia** + symmetric **proximal muscle** weakness
- Labs: increased serum **creatine kinase**
- Treatment: immunosuppressive agents

Dermatomyositis: Antibody-driven autoimmune disease characterized by **perimysial inflammation** and atrophy, resulting in muscle tissue damage; associated with **CD4** and complement activation

Polymyositis: T-cell driven autoimmune disease characterized by **endomysial inflammation**; associated with **CD8**



https://yalehistology.s3.us-east-2.amazonaws.com/muscle/skeletal_muscle_endomysium.jpg

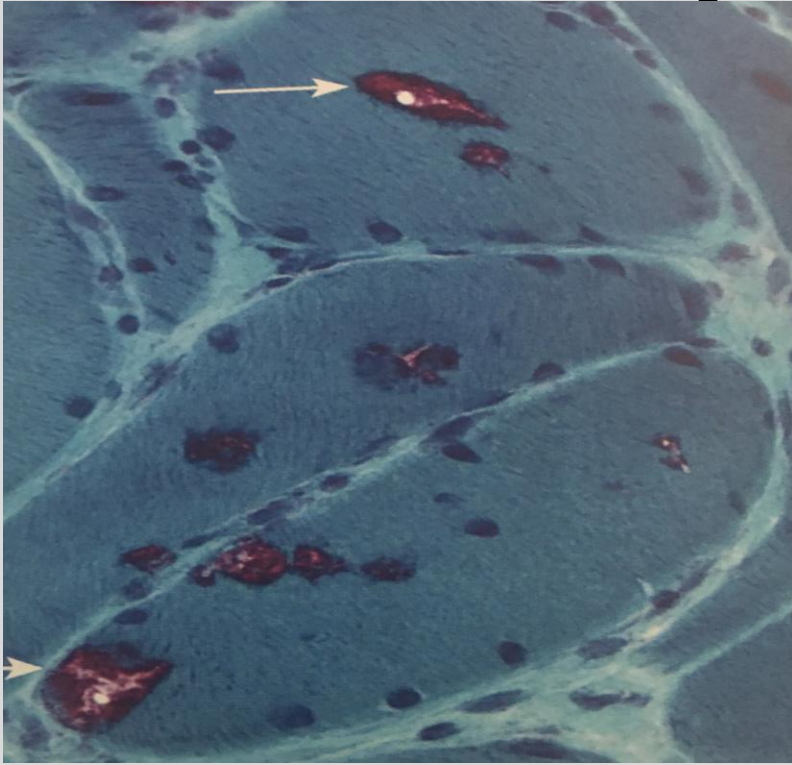


<https://www.ncbi.nlm.nih.gov/pubmed/>

Inclusion-Body Myositis

- **Most common inflammatory myopathy of patients age > 65yo**
- Pathogenesis: Unclear if inflammatory or degenerative process
- Clinical Findings:
 - Begins with **distal muscle involvement**, may be **asymmetric** (v. dermatomyositis and polymyositis)
 - Slowly progressive muscle weakness most severe in **quadriceps and distal upper extremities**, may have dysphagia

Inclusion Body Myositis



Gomori trichrome stain

Diagnosis/Labs:

- Moderately increased creatine kinase levels
- Can be **cytosolic 5'-nucleotidase 1A (cN1A) antibody positive (50% of patients)**
- Histology: **“rimmed vacuoles”** (autophagic vacuoles containing degrading organelles and other cytoplasmic contents) **in myofibers**
- CD8+ T cell endomysial infiltration similar to polymyositis

Treatment: Immunosuppressive agents or IVIG (**not as effective as in other inflammatory myopathies**)

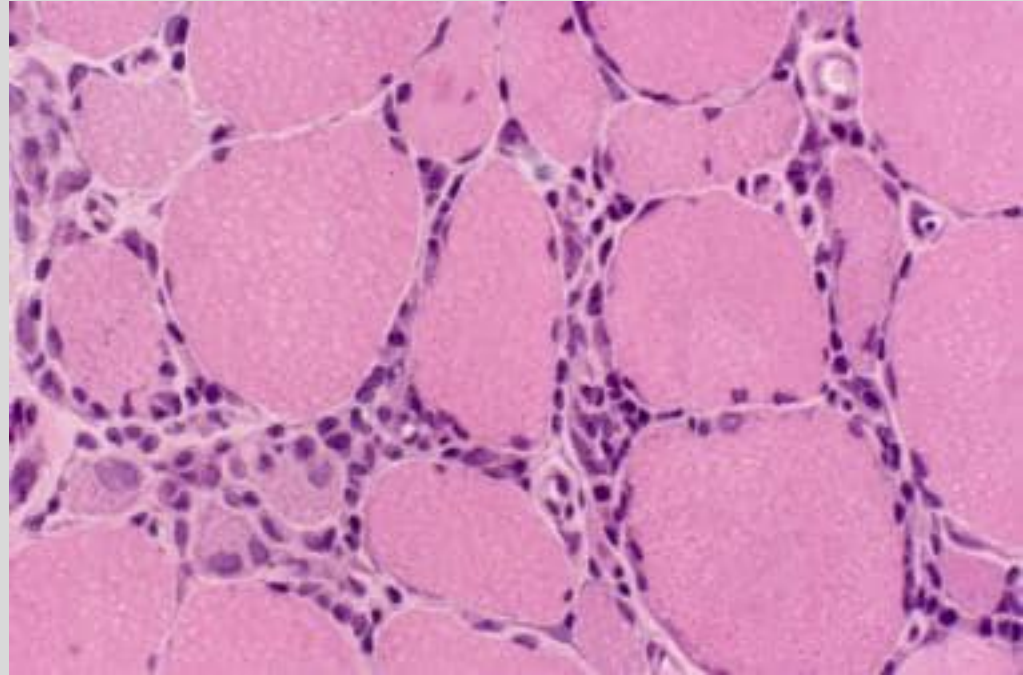
Robbins and Cotran, Pathologic Basis of Disease, 10th ed 2020

INFLAMMATORY MYOPATHIES

- DX based on
 - clinical symptoms
 - EMG (mixed neurogenic and myopathic findings suggestive of inflammatory myopathy)
 - increased CK (serum)
 - biopsy

Question

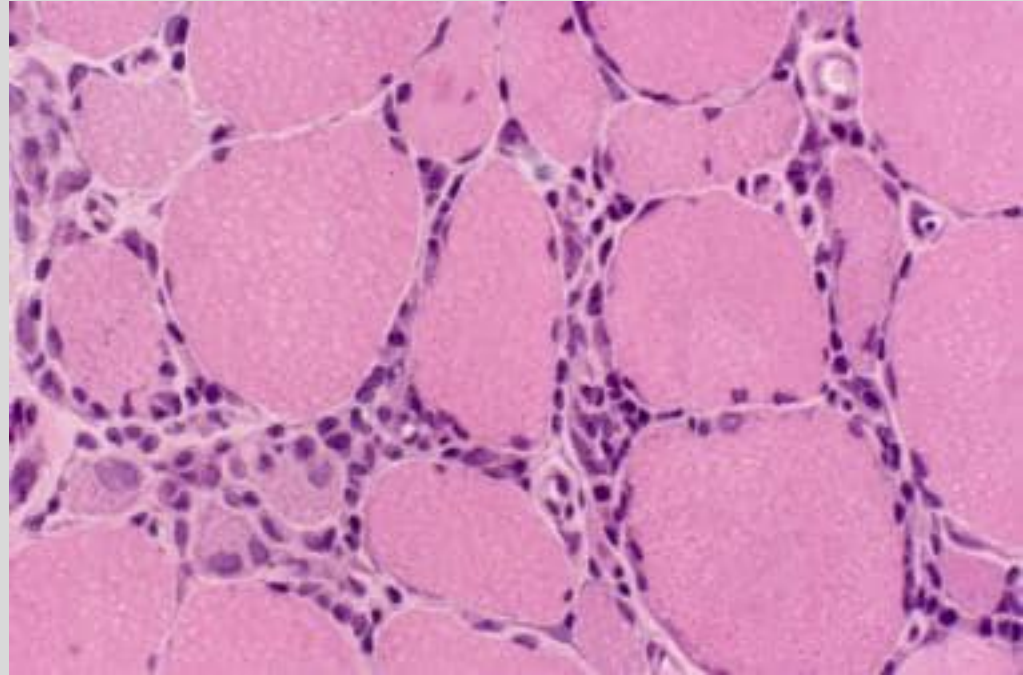
A 35-year-old female with no past medical history presents for her yearly physical. She mentions difficulty with washing her hair and ascending stairs. This has been happening for approximately 6 months. She has no other complaints. Her physician orders a muscle biopsy which shows endomysial inflammation and lymphocytes. What is the diagnosis?



- A. Dermatomyositis
- B. Inclusion Body Myositis
- C. Myasthenia Gravis
- D. Grave's Disease
- E. Polymyositis

Answer

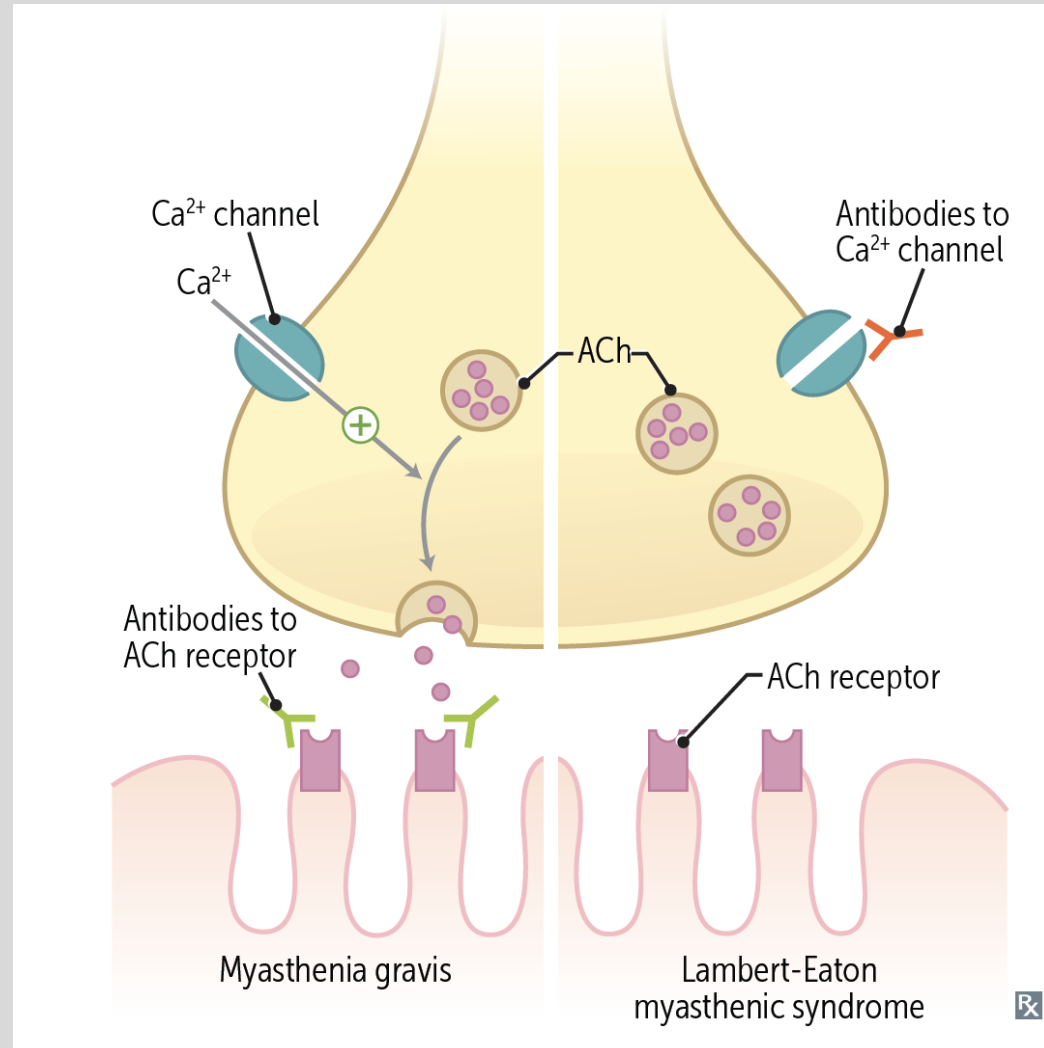
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- A. Dermatomyositis
- B. Inclusion Body Myositis
- C. Myasthenia Gravis
- D. Grave's Disease
- E. **Polymyositis**

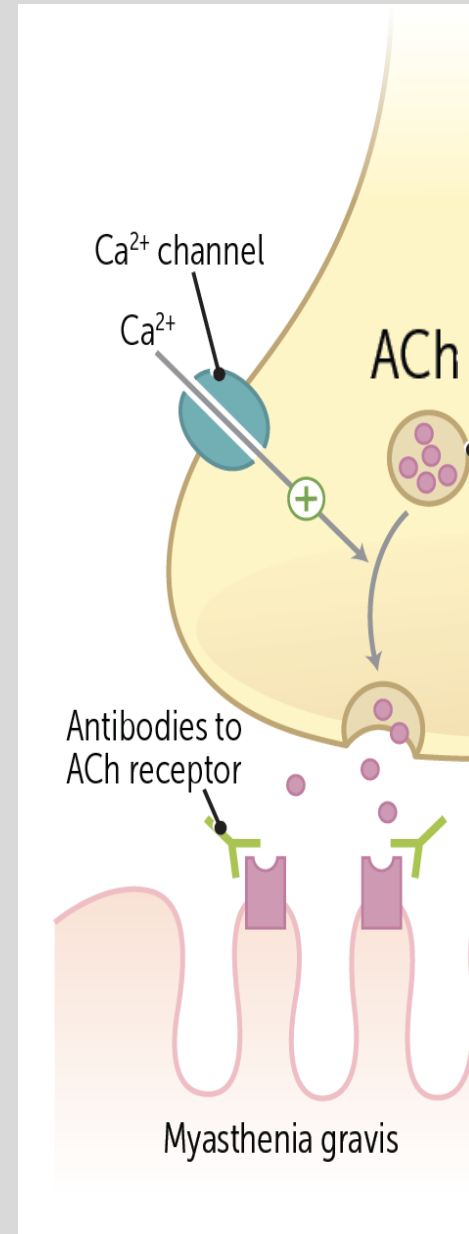
Diseases of the Neuromuscular Junction (NMJ)

- Myasthenia Gravis
- Lambert Eaton Myasthenic Syndrome



Myasthenia Gravis

- **Autoimmune - immune mediated loss of acetylcholine receptors**
- Pathogenesis: **Auto-abs** against **post-synaptic acetylcholine receptors** (~85%), against muscle acetylcholine receptor (**AChR-MG**) - leads to **destruction of receptors** and **limited response of myofibers to ACh**
- Pathologic mechanisms
 - complement fixation-mediated postsynaptic membrane damage
 - postsynaptic membranes show morphologic alterations, depletion of ACh-r leads to limited myofiber response to ACh

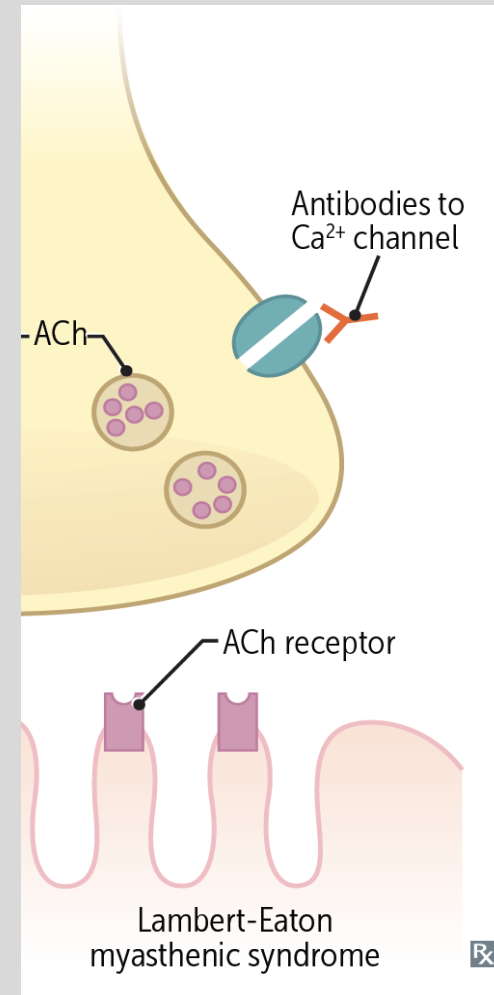


Myasthenia Gravis

- Clinical features:
 - Bimodal distribution; young **females (20-40 years)**, older **males (50-70 years)**, more commonly affects women (2:1), if arising before age 40 years
 - **Painless weakness, fatigue—worsens with repeated exertion**
 - Weakness in proximal muscles, diaphragm, neck extension
 - **Diplopia, ptosis**
 - Dysphagia for both solids and liquids
 - Reflexes spared
 - **Thymic abnormalities associated** (thymoma, thymic hyperplasia)—esp in younger pts (10% of patients have a thymoma, 30% of patients (usually younger) have thymic hyperplasia)
 - **Electrophysiologic studies** show decreased muscle response with repeated stimulation
 - **Nerve conduction** studies normal
- Treatment: **Acetylcholinesterase inhibitors** are first line (inhibit acetylcholinesterase which increases acetylcholine and reverses muscle weakness)
 - Others: plasmapheresis, immunosuppressive drugs, thymectomy (if thymoma)

Lambert Eaton Myasthenic Syndrome

- **Rare** autoimmune disease, more common in **males**, onset age peaks at **35 years**, with a larger peak at **60 years**
- Pathogenesis: autoantibodies against **pre-synaptic calcium channels that prevents the release of ACh**
- Clinical features: Muscle weakness that **improves with exertion**
 - **Autonomic symptoms** (dry mouth, constipation, impotence)
 - Hyporeflexia
 - 50% of patients have an **associated malignancy** (**neuroendocrine lung carcinoma, e.g., small cell carcinoma**)
 - Patients may have other autoimmune diseases as well
- Diagnosis: electrophysiologic studies
 - Repetitive stimulation increases the muscle response (**opposite of MG**)
- Treatment:
 - Drugs that increase the release of ACh into the synapse
 - Immunosuppressive agents



Question

- 24 yo female presents with new onset **blurry vision, worse at the end of the day**. Exam shows **drooping of the bilateral eyelids**. EMG findings are significant for **diminished muscle response after repeated stimulation**. What is the mechanism behind this patient's disease?
- A. antibodies against post-synaptic Ach-r
- B. antibodies against pre-synaptic Ca^{2+} channels
- C. immune-mediated endomysial inflammation
- D. Ryanodine receptor mutation

Answer

- 24 yo female presents with new onset **blurry vision, worse at the end of the day**. Exam shows **drooping of the bilateral eyelids**. EMG findings are significant for **diminished muscle response after repeated stimulation**. What is the mechanism behind this patient's disease?
- **A. antibodies against post-synaptic Ach-r**
- B. antibodies against pre-synaptic Ca^{2+} channels
- C. immune-mediated endomysial inflammation
- D. Ryanodine receptor mutation

Summary Slide 1

- **SMA1 (Werdnig-Hoffman disease)** - autosomal recessive; degeneration of anterior horns (motor neurons) of spinal cord; loss-of-function mutation in SMN1 gene causes defects, “floppy” baby (hypotonia)
- **Duchenne muscular dystrophy** - X-linked recessive, frameshift deletion or nonsense mutation of dystrophin gene, **absent dystrophin protein**; early childhood, increased CK; proximal muscle weakness and clumsiness; **pseudohypertrophy of calf muscles**: (“fibrofatty infiltration”), **Gower’s sign**, cardiomyopathy, arrhythmias, intellectual impairment
- **Becker muscular dystrophy** - X-linked recessive, loss-of-function mutation of dystrophin gene results in truncated, **partially functional dystrophin protein, less severe**, slower progression than DMD
- **Myotonic dystrophy** - autosomal dominant, **CTG trinucleotide repeat** in DMPK gene, **CLC1 deficiency**, myotonia, muscle wasting, cataracts, frontal balding, testicular atrophy, etc.
- **Malignant hyperthermia** - autosomal dominant with variable penetrance; mutation ryanodine receptor (RYR1) gene results in increased calcium release from sarcoplasmic reticulum triggered by halogenated inhaled anesthetics and succinylcholine: hypermetabolic state (hyperthermia, muscle spasms, tachycardia, etc.)
- **Congenital myopathies- Central Core Disease** (AD, mutation of RYR1 gene, early onset hypotonia, skeletal abnormalities, cytoplasmic cores)/ **Nemaline Myopathy** (AD or AR, NEM 1-7 gene mutations; childhood weakness/“floppy” baby, Nemaline rods on biopsy)

Summary Slide 2

- **Diseases of lipid metabolism - Carnitine Palmitoyltransferase II Deficiency;** impaired transport of **free fatty acids** into mitochondria: build up of fatty acids in cytosol, episodic muscle damage with exercise or fasting, hypotonia, hypoketotic hypoglycemia, dilated cardiomyopathy
- **Diseases of glycogen metabolism:**
 - **McArdle disease** (Myophosphorylase deficiency (type V), **episodic muscle damage with exercise, painful muscle cramps, myoglobinuria, normal glucose**)
 - **Pompe disease** (acid maltase deficiency type II; impaired glycogen breakdown within lysosome; cardiomegaly/hypertrophic cardiomyopathy, hypotonia, exercise intolerance)
- **Mitochondrial myopathies** -mitochondrial inheritance (maternal transmission); mutations **impair ATP generation** in mitochondria; myopathy, lactic acidosis & CNS disease; MELAS syndrome; fatigue, weakness, rhabdomyolysis, chronic progressive external ophthalmoplegia, elevated CK

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>